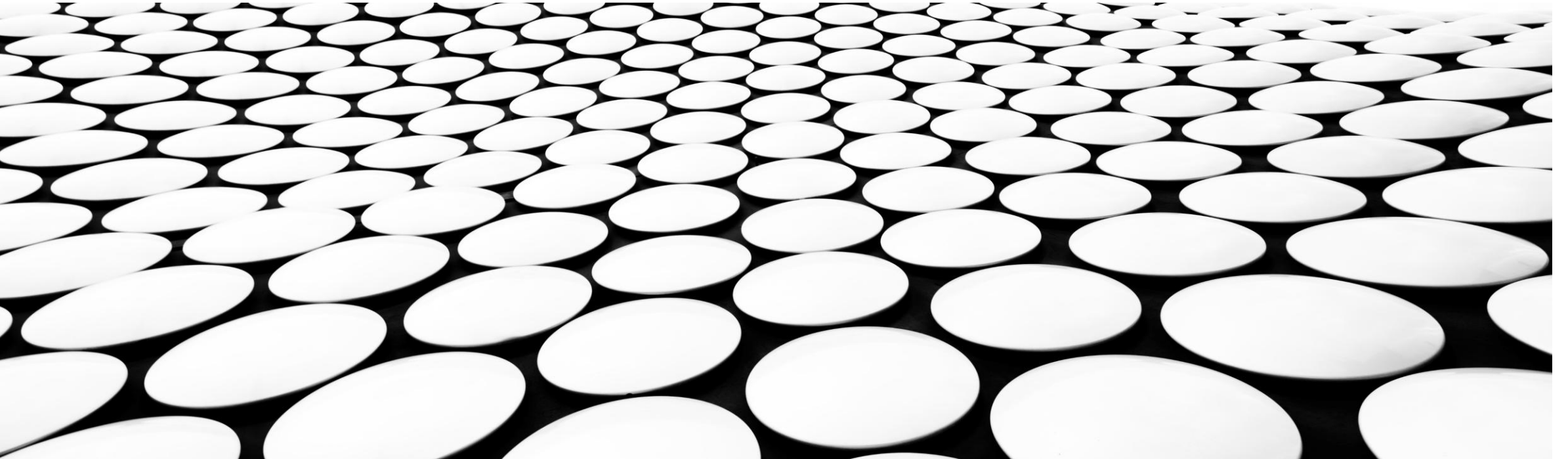

SMART MOVIE RECOMMENDATION SYSTEM

MACHINE LEARNING PROJECT



PROJECT OVERVIEW

- Developed a Content-Based Movie Recommendation System.
- Recommends similar movies based on the user's selected movie.
- Uses Machine Learning and Natural Language Processing (NLP).
- Built with Python and Streamlit.



PROBLEM STATEMENT

- Finding movies according to user interests can be time-consuming.
- This project helps users discover similar movies quickly using machine learning techniques based on movie content.

TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Scikit-learn
- NLTK
- Streamlit
- Pickle

DATASET

➤ Dataset

TMDB 5000 Movie Dataset

➤ Files Used:

tmdb_5000_movies.csv

tmdb_5000_credits.csv

➤ Dataset includes:

☐ Movie Title

☐ Genres

☐ Cast

☐ Crew

☐ Keywords

☐ Overview

PROJECT WORKFLOW

- Data Collection
- Data Cleaning
- Feature Engineering
- Text Vectorization
- Cosine Similarity
- Recommendation Generation
- Streamlit Deployment

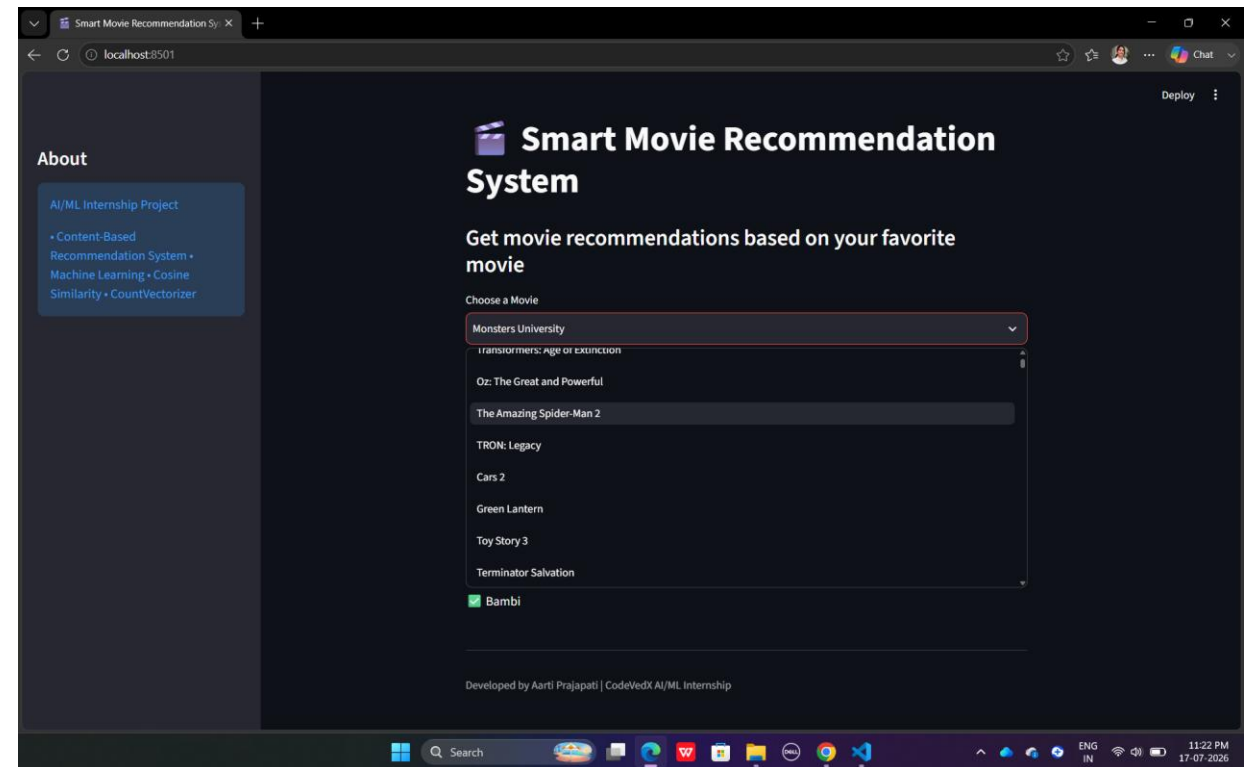
FEATURES

- Movie Selection Dropdown
- Top 5 Movie Recommendations
- Content-Based Filtering
- Fast Recommendation Generation
- User-Friendly Streamlit Interface

RESULTS

Project Output

- Successfully recommends similar movies.
- Interactive Streamlit application.
- Machine Learning model saved using Pickle.
- Accurate recommendations using Cosine Similarity.



CONCLUSION

- Successfully built a Content-Based Recommendation System.
- Improved understanding of Machine Learning and NLP concepts.
- Learned data preprocessing, feature engineering, vectorization, and model deployment using Streamlit.



THANK YOU!